

QPROG Project

QUBOs for Combinatorial Problems

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1 General Structure of the Project

1. Choose an NP-hard graph theoretic problem. See a short list in Appendix A.
2. Provide a QUBO formulation for the problem. 10%
3. Implement a python program that takes as input a graph and solves the problem using a local QAOA simulator. 25%
4. Implement a python program that takes as input a graph and solves the problem using a quantum computer on the cloud. This could be using any provider of Quantum Computing Services. See a list in Appendix B. 25%
5. Implement a brute-force solver for the problem. 10%
6. Build a dataset containing 10 graphs of up to 50 vertices. Make an experimental evaluation of the performance of the local simulator, the cloud simulator and the brute force implementation. 30%
7. **Bonus:** solving your QUBO on a non-IBM quantum computer on the cloud is a bonus problem. 10%
8. **Observation:** Please feel free to base your implementation on standard tutorials available online. However two things should be observed:
 - You should add a link to the tutorial you used in your report.
 - The combinatorial problem you choose should be significantly distinct from the one used in the tutorial. For example, if the tutorial deals with the Hamiltonian path problem, do not choose the Hamiltonian path problem nor the traveling salesman problem since these problems are too closely related.

A Some NP-Hard Problems

1. **Vertex Cover:** Find the smallest subset of vertices such that every edge in the graph is incident to at least one vertex in the subset.
2. **Clique Problem:** Determine whether a graph contains a clique (a subset of vertices where every two distinct vertices are adjacent) of a specified size.
3. **Graph Coloring:** Assign colors to the vertices of a graph such that no two adjacent vertices share the same color, using the minimum number of colors possible.
4. **Hamiltonian Cycle:** Decide whether a graph contains a cycle that visits each vertex exactly once.
5. **Maximum Independent Set** Find the largest set of vertices in a graph, no two of which are adjacent.
6. **Steiner Tree Problem** Given a set of terminal vertices, find the minimum-weight tree that spans all the terminals, possibly including additional non-terminal vertices.
7. **Dominating Set:** Find the smallest set of vertices such that every vertex in the graph is either in the set or adjacent to at least one vertex in the set.

B Some Providers of Quantum Computing Services

1. D-Wave Leap
2. IBM Quantum Experience
3. Amazon Braket
4. Azure Quantum
5. Quantum Cloud Services (QCS)
6. IonQ
7. Strawberry Fields
8. QC Ware
9. Zapata Computing
10. Quantinuum Quantum Cloud